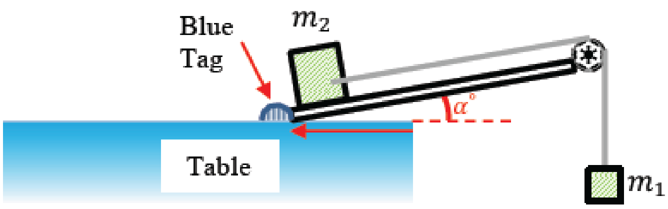
Tutorial 2
Newton's Law of Motion & Forces

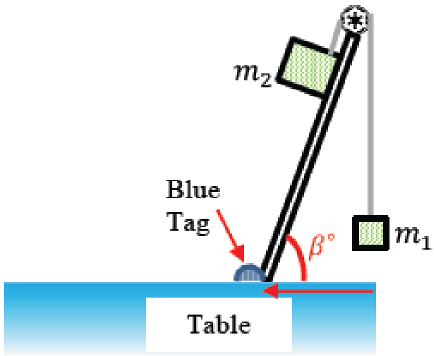
1. Static Friction

You are given a rough ramp with coefficient of static friction μ_s, a smooth pulley and two different masses m_1 and $m_2(>m_1)$ connected by a masses string. The setup is as shown in (A). When the ramp is held at an angle α, the block with mass m_2 just starts to slide up the ramp. By increasing the steepness of the ramp to a certain angle β as shown in (B), the block with mass m_2 just starts to slide down the ramp.

(A)



(B)



Show that

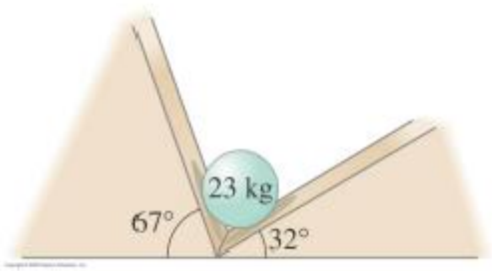
(i) From diagram (A) we have $\mu_s = \frac{m_1 - m_2 \sin\alpha}{m_2 \cos\alpha}$.

(ii) From diagram (B) we have $\sqrt{1+\mu_s^2}\sin(\beta - \tan^{-1}\mu_s) = \frac{m_1}{m_2}$. Hint: Consider the R-formula.

(iii) The difference in angles is given by $\beta - \alpha = 2\tan^{-1}(\mu_s)$

2. Contact Forces

[G12.73] A 23-kg sphere rests between two smooth planes as shown in Figure G12.73. Determine the magnitude of the forces acting on the sphere exerted by each plane.



G12.73

Answer: $F_L \approx 120N; F_R \approx 210\ N$

3. **Drag Force – Streamline flow: Stoke's law**

Consider an object released from rest in a liquid. Ignoring upthrust, it is subjected to the drag force $F_D = bv$ and its weight $W = mg$. Its velocity increases from zero to the terminal velocity and is described at all times by the expression:

$$v(t) = \frac{mg}{b}\left(1 - e^{-\frac{bt}{m}}\right)$$

Assuming $s=0$ when $t=0$

a) Derive an equation for the displacement $s(t)$ of the object. {Answer: $s(t) = \frac{mg}{b}t + \frac{m^2g}{b^2}e^{-\frac{bt}{m}} - \frac{m^2g}{b^2}$ }

b) Calculate the displacement of the object when it reaches 30% of terminal velocity.

{Answer: $s = \frac{m^2g}{b^2}[0.7 - \ln 0.7) - 1]$ $= -\frac{m^2g}{b^2}[\ln(0.7) + 0.3]$

c) When the object reaches 30% of its terminal velocity, what percentage of its maximum KE is reached? {Answer: 9%}

4. **Drag Force – Turbulence flow**

['A' J86/II/8] When a body moves through a fluid, a retarding force due to turbulence may be experienced. In the case of a sphere of radius r moving with speed v through a stationary fluid of density ρ which is at rest, this force is

$$F = k\rho\, r^2 v^2$$

where k is a constant.

When spherical raindrops fall through still air, all but the smallest experience a retarding force given by the equation above. It is found that drops of a given radius approach the ground with an approximately constant speed, which is independent of the cloud in which they are form.

(a) Find an expression for this terminal speed v_t in terms of the constant k , the radius r of the drop, its density ρ_w , the density of air ρ_A of the air and the acceleration of free fall g. (You may neglect the upthrust due to air). Sketch the velocity-time graph of the object from the moment it is released into the fluid.

(b) The terminal speed of the raindrop of radius 1 mm is approximately 7 ms^{-1}. In freak storms, hailstones with radii as large as 20 mm may fall. Estimate the speed with which such stones strike the ground. [Take the density of water as $1 \times 10^3\ kg\ m^{-3}$ and the density of ice as $9 \times 10^2\ kg\ m^{-3}$.

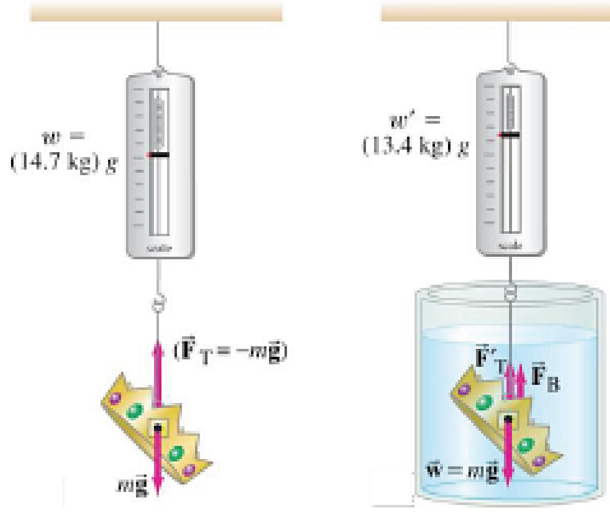
Answer:

(a) $v_t = \sqrt{\frac{4\rho_w \pi r g}{3k\rho_A}}$ (b) 30 m/s

5. Upthrust

a) The density of ice is 0.917 g/cm³, whereas that of seawater is 1.025 g/cm³. What percent of an iceberg is above the surface of the water? (Discussion) If some ice and seawater are placed in a measuring cylinder, will the seawater level rise, drop or stay the same when all the ice melts?

b) When a crown of mass 14.7 kg is submerged in water, an accurate scale reads only 13.4 kg. Is the crown made of gold?

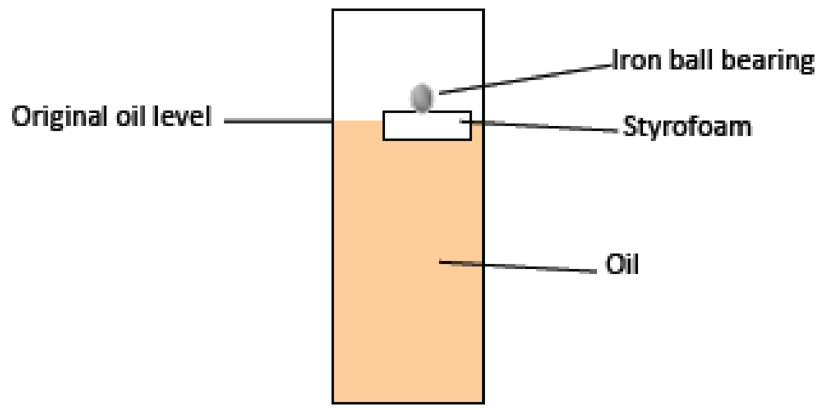


c) A spherical iron ball bearing of diameter 4 mm is placed on a piece of Styrofoam floating in a cylinder of oil as shown in the figure below. You may assume that the weight of the Styrofoam is negligible. Density of iron is 7.86 g cm^{-3}; density of oil is 0.83 g cm^{-3}.

i) Calculate the upthrust acting on the piece of Styrofoam.

ii) What is the volume of oil displaced by the piece of Styrofoam and the iron ball bearing?

iii) The ball bearing rolls off the piece of Styrofoam and falls into the oil. Will the oil level be higher, lower or the same as before? Explain your answer.



Answer:

(a) 10.5 %, seawater level rises.

(b) Not made of gold.

(c)(i) $2.58 \times 10^{-3} N$ (ii) $0.317\ cm^3$ (iii) Lower

6. Static and Kinetic Friction

A 28.0-kg block is connected to an empty 2.00-kg bucket by a cord running over a frictionless pulley (Fig. 52). The coefficient of static friction between the table and the block is 0.45 and the coefficient of kinetic friction between the table and the block is 0.32. Sand is gradually added to the bucket until the system just begins to move. (*a*) Calculate the mass of sand added to the bucket. (*b*) Calculate the acceleration of the system.

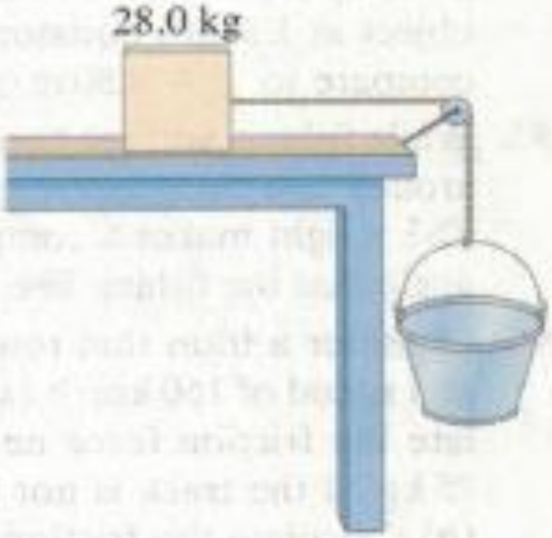
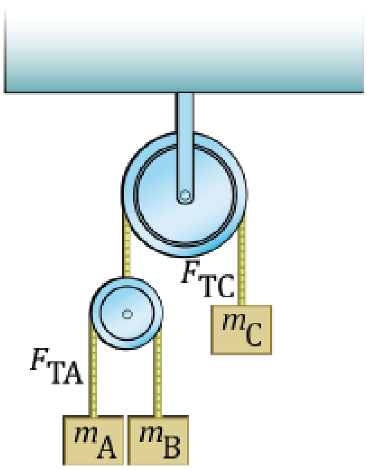


FIGURE 52
Problem 88.

Answer: 11 kg; 0.88 m/s^2

7. Relative Pulleys

Dynamics Problems >> Giancoli problem 4.56:



The double Atwood machine has frictionless, massless pulleys and cords. Determine

a) the acceleration of masses m_A, m_B and m_C, and
b) the tensions F_{TA} and F_{TC} in the cords.
(Hint: You need to think about relative acceleration of m_A and m_B with respect to the movable pulley in order to relate all 3 accelerations.)

Answer:
(a) Depend on the direction of acceleration drawn.
(b) $F_{T_A} = \frac{4m_A m_B m_C}{4m_A m_B + m_B m_C + m_A m_C} g$; $F_{T_C} = \frac{8m_A m_B m_C}{4m_A m_B + m_B m_C + m_A m_C} g$

8. **Moving Pulleys**

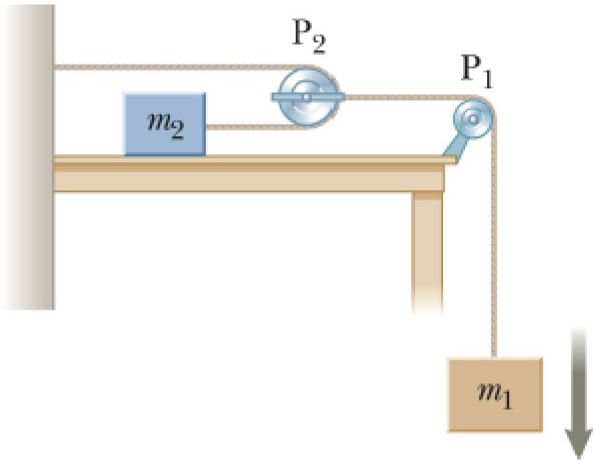
[S5.34] An object of mass m_1 hangs from a string that passes over a very light fixed pulley P_1 as shown in Figure S5.34. The string connects to a second very light pulley P_2. A second string passes around this pulley with one end attached to a wall and the other to an object of mass m_2 on a frictionless, horizontal table.

(a) If a_1 and a_2 are the accelerations of m_1 and m_2, respectively, what is the relation between these accelerations?

Find expressions for

(b) the tensions in the strings and

(c) the accelerations a_1 and a_2 in terms of the masses m_1 and m_2, and g.



S5.34

Answer:

(a) $a_2 = 2\,a_1$, (b) $T_1 = m_1\,g - m_1\,a_1$, $T_2 = m_2\,a_2$, (c) $a_1 = \frac{m_1 g}{m_1 + 4\,m_2}$, $a_2 = \frac{2 m_1 g}{m_1 + 4 m_2}$

9. **Newton's Laws of Motion**

Consider a ship of mass 2×10^6 kg with propeller blades of radius 7 m and it can push water backwards to a speed of 2 m/s. Given that the density of water is 1000 kg/m³, find the acceleration of the ship. {Answer: 0.308 m/s^2}

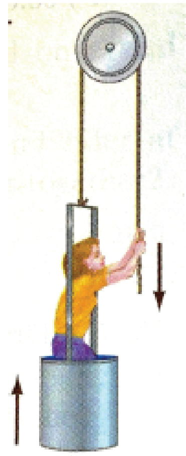
10. Newton's Laws of Motion

A window washer pulls herself upward using the bucket-pulley apparatus as shown in the figure. The mass of the person plus the bucket is 72kg.

a) How hard must she pull downward to raise herself slowly at constant speed?

b) If she increases this force by 15%, what will her acceleration be?

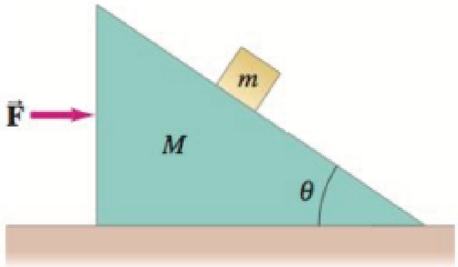
{Answer: a) 350N, b) 1.5 m/s^2}



11. Newton's Laws of Motion

A small block of mass m rests on the sloping side of a triangular block of mass M which itself rests on a horizontal table as shown in the figure. Assuming all surfaces are frictionless, determine an expression for the force $\vec{F}$ that must be applied to M so that m remains in a fixed position relative to M (that is, m doesn't move on the incline). [Hint: Take x and y axes horizontal and vertical.]

{Answer: $F = (m + M)g \tan \theta$}



12. Newton's Laws of Motion

In Figure 1, a small block of mass m is sent sliding with initial velocity v along a slab of mass $10m$, starting at a distance of l from the far end of the slab. The coefficient of kinetic friction between the slab and the floor is μ_1; that between the block and the slab is μ_2, with $\mu_2 > 11\,\mu_1$.

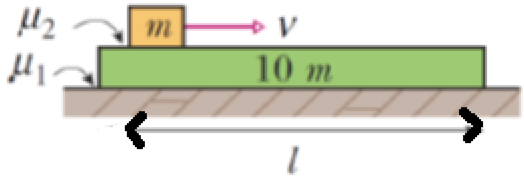


Figure 1

(a) Derive an expression for the minimum value of v such that the block reaches the far end of the slab.

(b) For the expression of v in part (a), derive an expression for the time the block takes to reach the far end.

Answer: $v_{min} = \sqrt{\frac{11}{5}(\mu_2 - \mu_1)gl}$; $t = \sqrt{\frac{20l}{11(\mu_2-\mu_1)g}}$

13. Simple Harmonic Motion

Two points A and B on a smooth horizontal table are at a distance 8l apart. A particle of mass m between A and B is attached to A by means of a light elastic string of modulus λ and natural length 2l, and to B by means of a light elastic string of modulus 4λ and natural length 3l. If M is the midpoint of AB, and O is the point between M and B at which the particle would rest in equilibrium, prove that MO = $\frac{2}{11}l$. If the particle is held at M and then released, show that it will move with simple harmonic motion and find the period of the motion. Find the velocity V of the particle when it is at a point C distant $\frac{3}{11}l$ from M, and is moving towards B. Recall Hooke's law:

$$T = kx = \lambda\frac{x}{a}$$

where T is the tension, k is the spring constant, x is the extension, λ is the string modulus and a is the natural length of the string.

Answer: $2\pi\sqrt{\frac{6ml}{11\lambda}}$; $\sqrt{\frac{\lambda l}{22m}}$

14. Simple Harmonic Motion

One end of an elastic string of natural length 1m is attached to a fixed point and the other end is attached to a particle of mass m kg. When the particle hangs in equilibrium the extension of the string is 0.05 m. The particle is pulled down a further 0.1 m and released from rest. Taking the value of g to be 9.8 m/s^2, find the time that elapses before the string becomes slack and the speed of the particle at that instant.

Prove that the time elapses between the instant of release and the first return of the particle to the point of release is $\frac{1}{21}(2\pi + 3\sqrt{3})$ seconds.

Answer: π/21 seconds; $\frac{7\sqrt{3}}{10}$ m/s